Experiment :-7

Problem Statement:-

Implement a sliding window protocol with piggybacking for efficient data transmission and error control. Simulate data transfer between two nodes and visualize the window movements and acknowledgments using Cisco Packet Tracer.

Aim:- Implementation of Sliding Window Protocol with Piggybacking

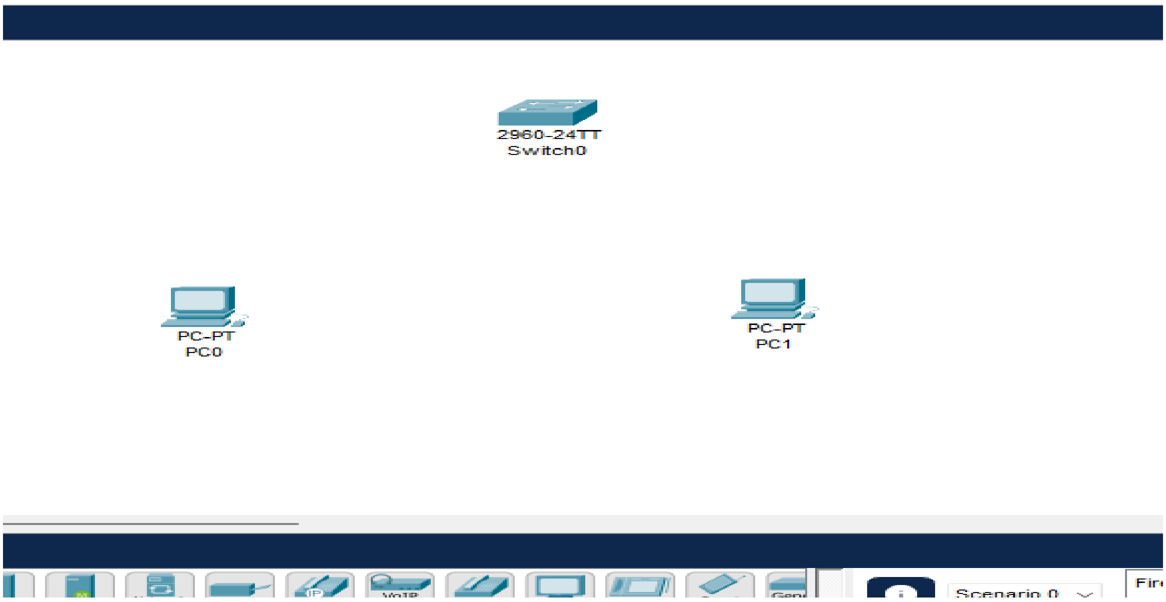
Procedure:-

Step 1: Open Cisco Packet Tracer

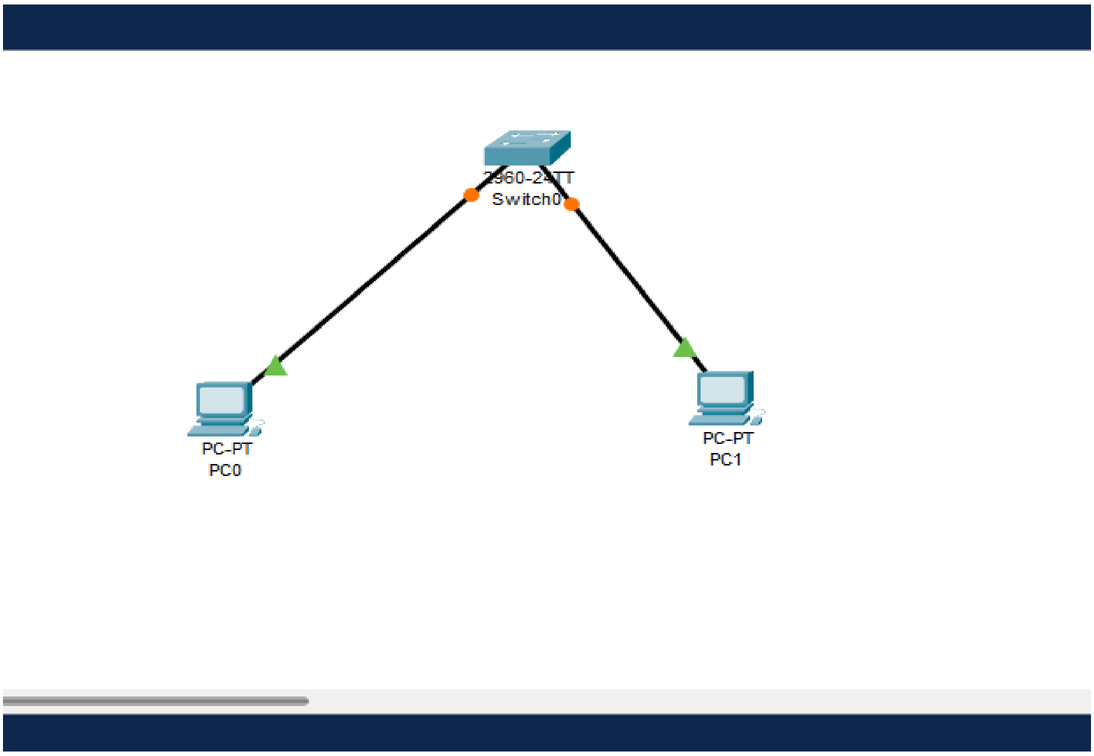
1. Launch Cisco Packet Tracer
2. You'll see a blank workspace in the middle

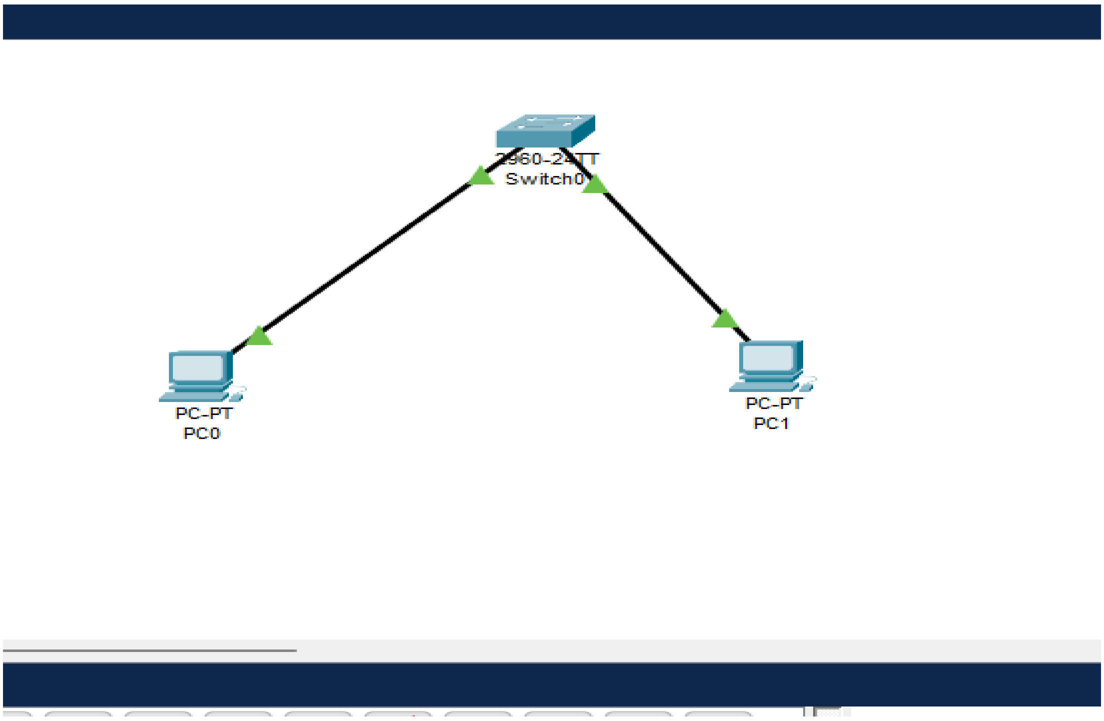
Step 2: Add Two PCs and one switch

1. At the bottom panel, click **"End Devices"** (PC icon)
2. Drag and drop **PC0** onto the workspace (left side)
3. Drag and drop **PC1** onto the workspace (right side)
4. Leave some space between them

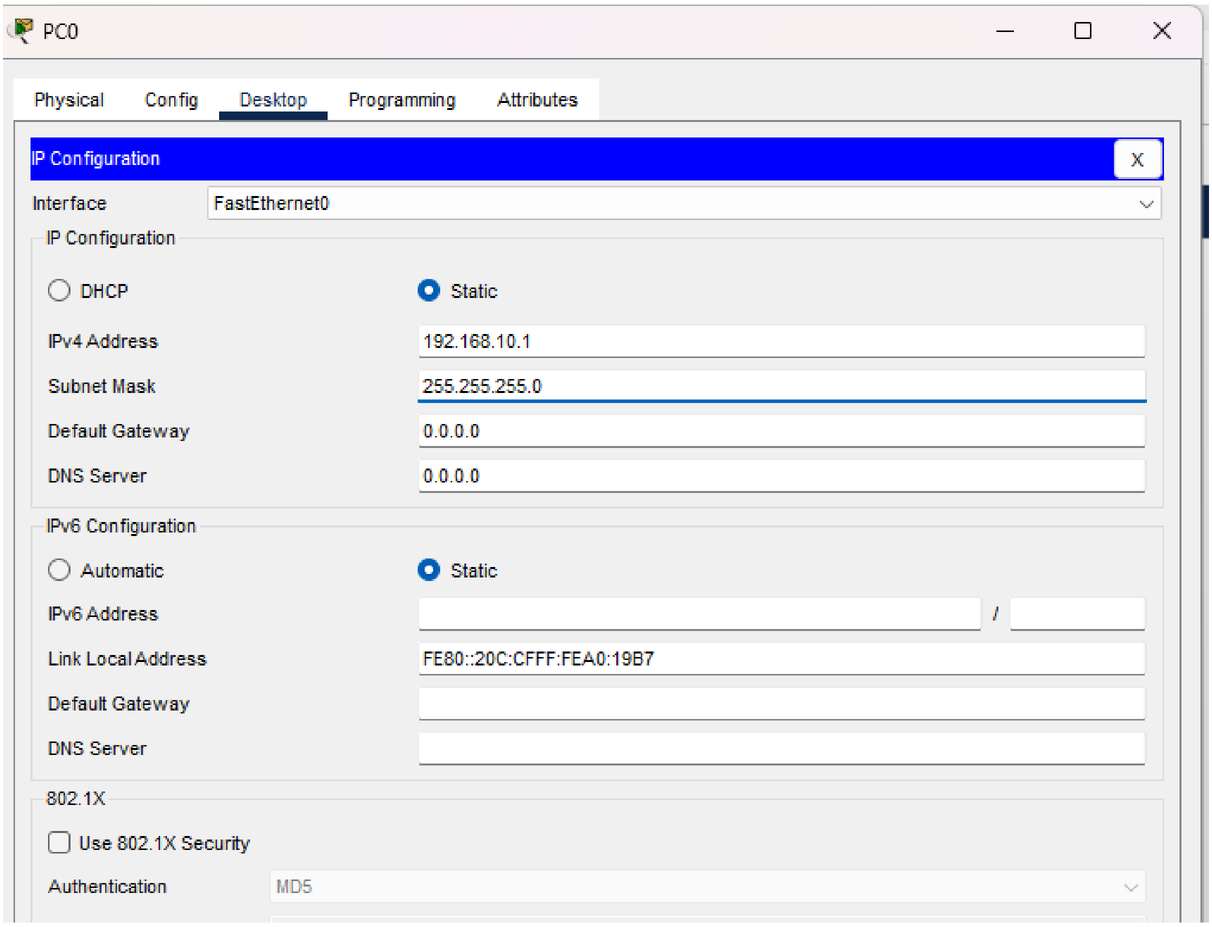


Step 3: Connect the PCs





Step 4: Assign IP to PC0



SLIDING WINDOW PROTOCOL:

- Used for Flow Control & Error Control
- Sender can send multiple frames before
  waiting for acknowledgment
- Window Size controls number of unacknowledged
  frames allowed at a time
- Improves network efficiency & throughput

PIGGYBACKING:

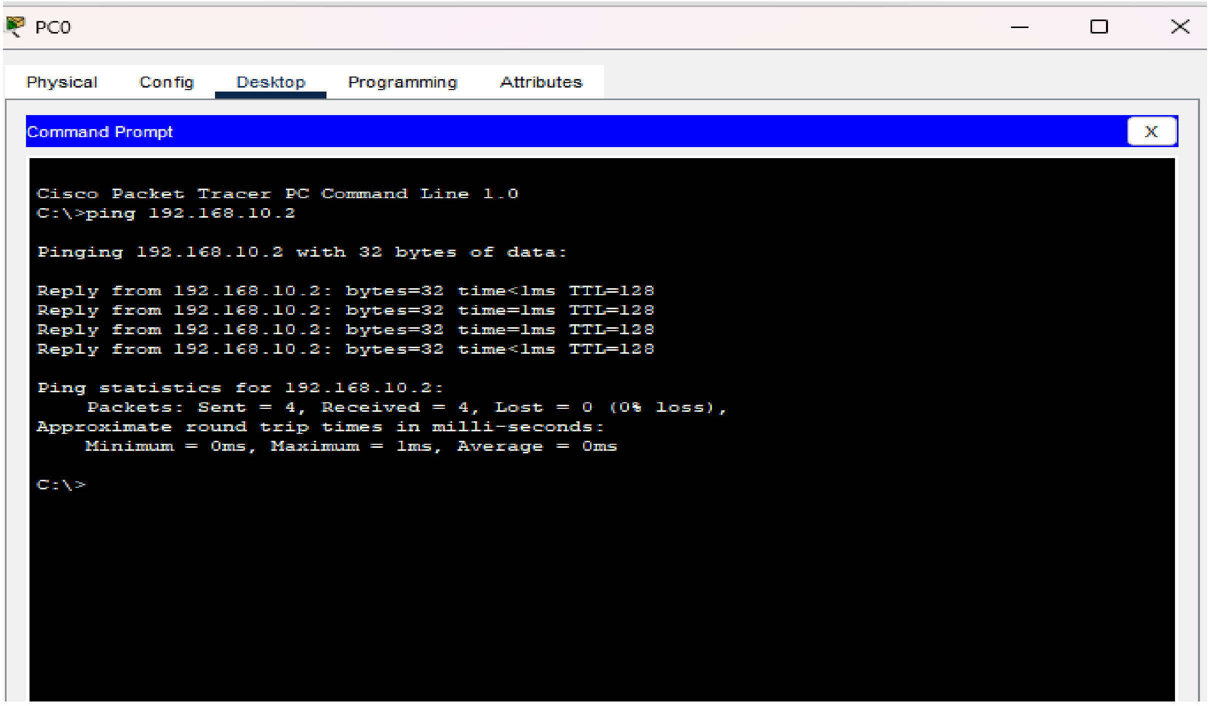
- Acknowledgments are attached/included
  inside outgoing DATA frames
- Avoids sending separate ACK packets
- Saves bandwidth and improves efficiency

- PC1 sends ACK inside its own data frames

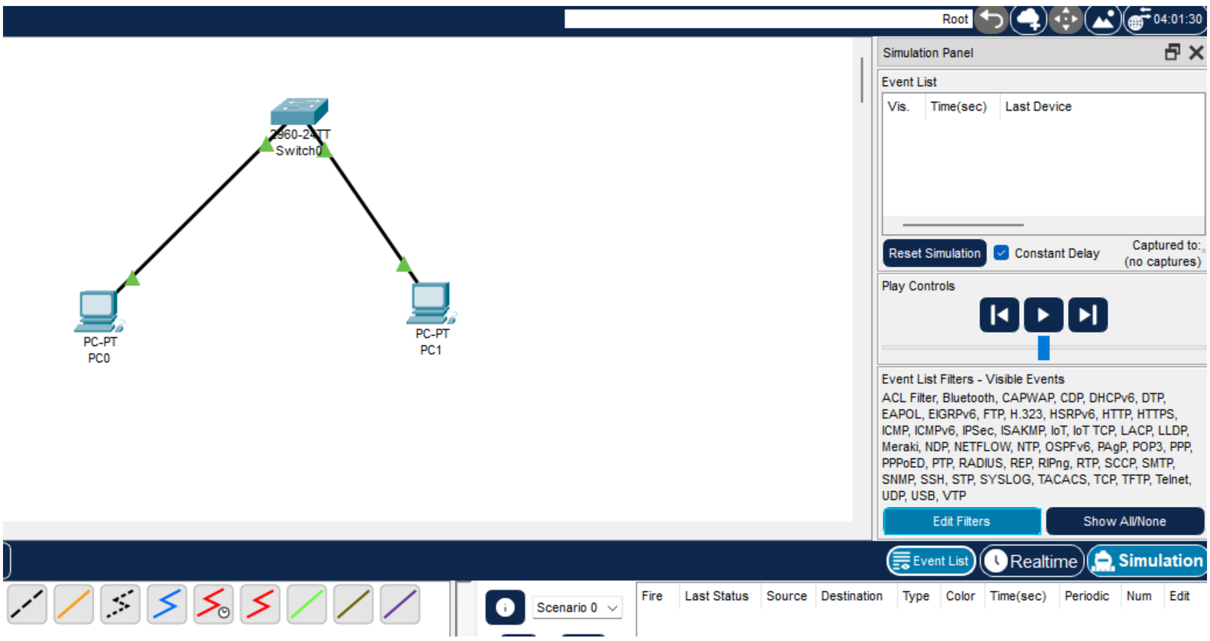
back to PC0

Step 8: Ping from PC0 to PC1

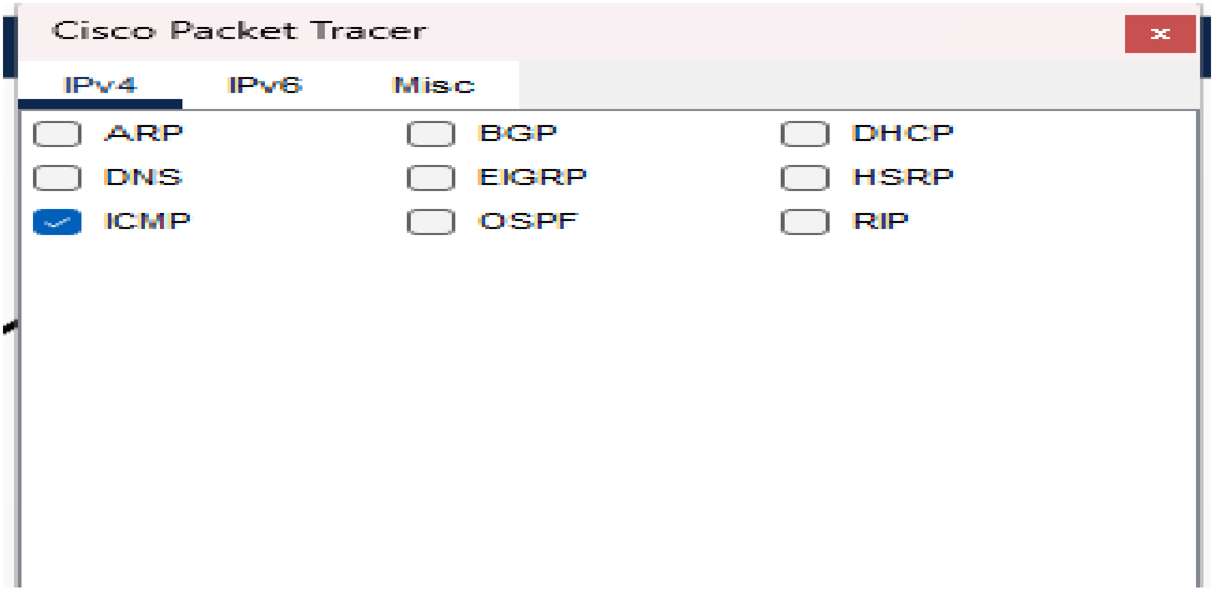
1. **Double-click PC0**
2. Go to **Desktop** tab → click **"Command Prompt"**



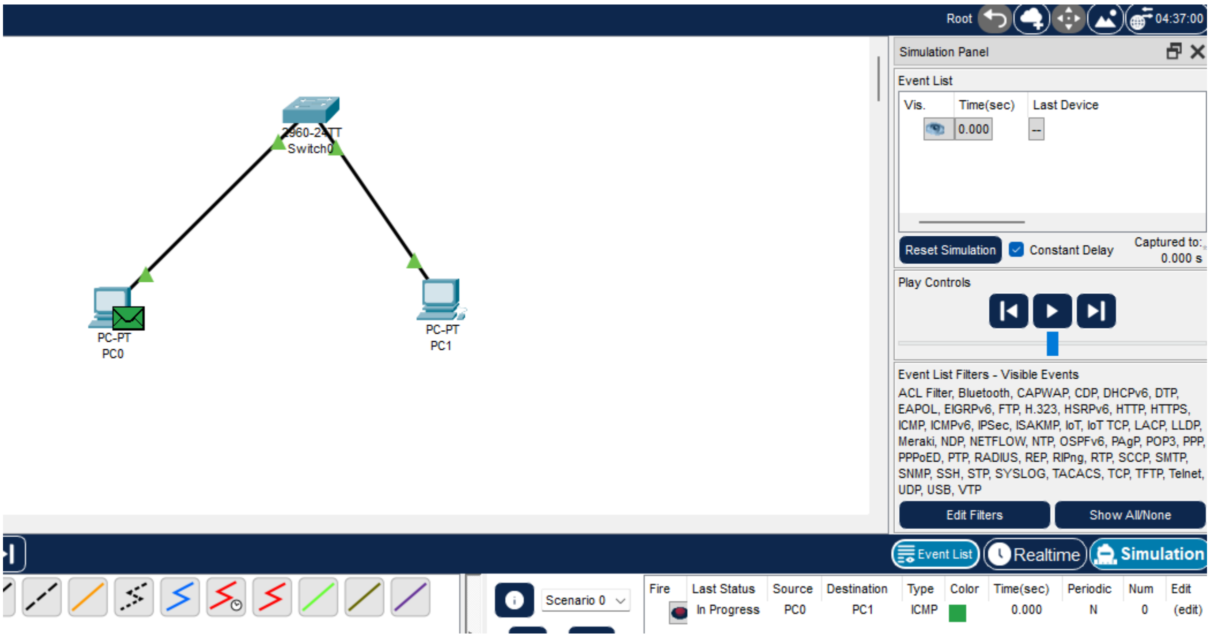
Step 9: Switch to Simulation Mode



Step 10: Filter Events

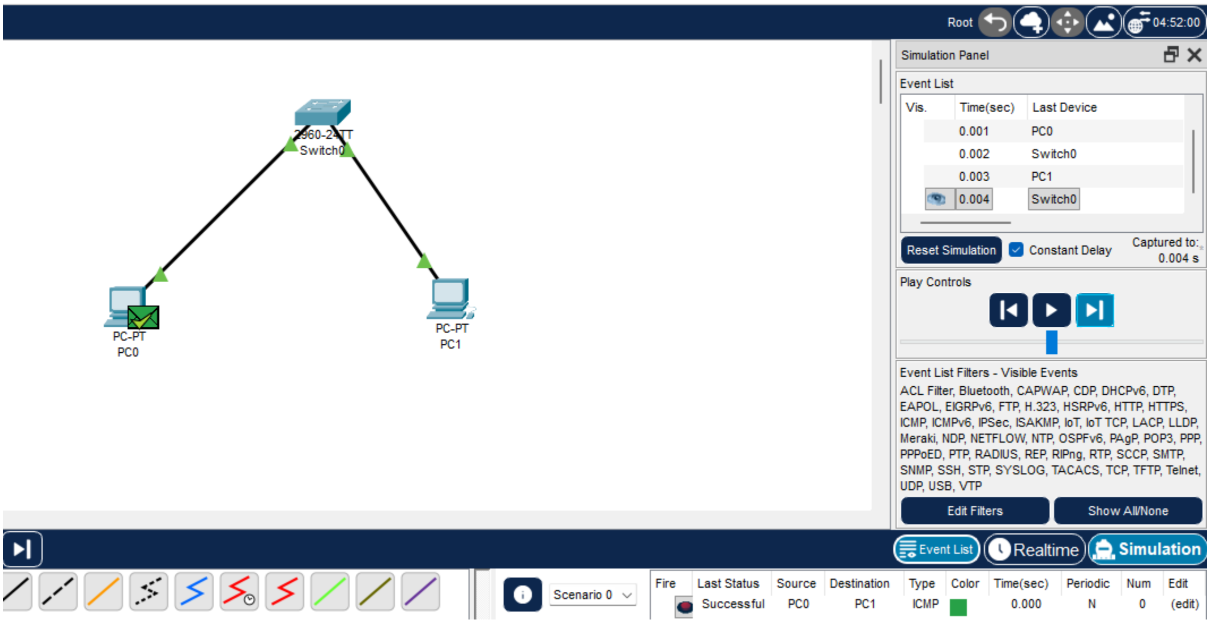


Step 11: Add Simple PDU (Data Frame PC0 → PC1)

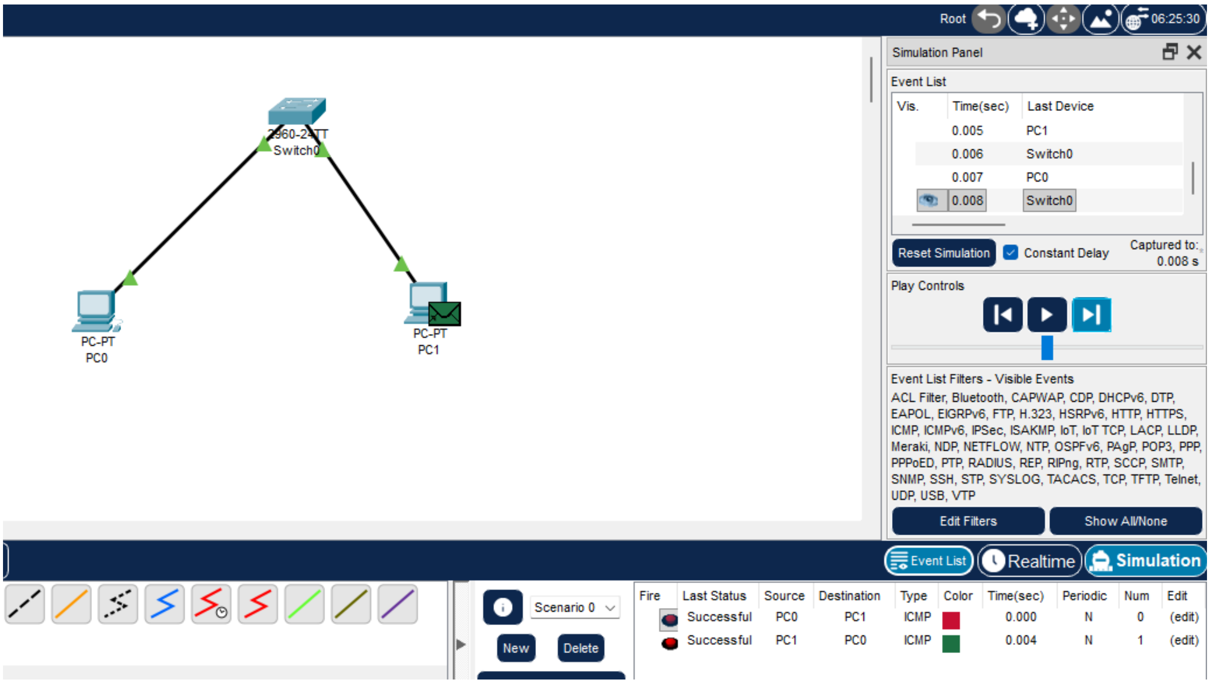


Step 12: Add Return PDU (Piggybacked ACK — PC1 → PC0)

1. Click **"Add Simple PDU"** again
2. **Click on PC1** first (source)
3. **Click on PC0** second (destination)
4. This simulates **piggybacking** — PC1 sends data back with ACK included



Root
04:52:00
2960-24TT
Switch0
PC-PT
PC0
PC-PT
PC1
Simulation Panel
Event List
Vis. Time(sec) Last Device
0.001 PC0
0.002 Switch0
0.003 PC1
0.004 Switch0
Reset Simulation
Constant Delay
Captured to:
0.004 s
Play Controls
Event List Filters - Visible Events
ACL Filter, Bluetooth, CAPWAP, CDP, DHCPv6, DTP, EAPOL, EIGRPv6, FTP, H.323, HSRPv6, HTTP, HTTPS, ICMP, ICMPv6, IPSec, ISAKMP, IoT, IoT TCP, LACP, LLDP, Meraki, NDP, NETFLOW, NTP, OSPFv6, PAgP, POP3, PPP, PPPoED, PTP, RADIUS, REP, RIPng, RTP, SCCP, SMTP, SNMP, SSH, STP, SYSLOG, TACACS, TCP, TFTP, Telnet, UDP, USB, VTP
Edit Filters
Show All/None
Event List
Realtime
Simulation
Scenario 0
Fire Last Status Source Destination Type Color Time(sec) Periodic Num Edit
Successful PC0 PC1 ICMP 0.000 N 0 (edit)



Root
06:25:30
2960-24TT
Switch0
PC-PT
PC0
PC-PT
PC1
Simulation Panel
Event List
Vis. Time(sec) Last Device
0.005 PC1
0.006 Switch0
0.007 PC0
0.008 Switch0
Reset Simulation
Constant Delay
Captured to:
0.008 s
Play Controls
Event List Filters - Visible Events
ACL Filter, Bluetooth, CAPWAP, CDP, DHCPv6, DTP, EAPOL, EIGRPv6, FTP, H.323, HSRPv6, HTTP, HTTPS, ICMP, ICMPv6, IPSec, ISAKMP, IoT, IoT TCP, LACP, LLDP, Meraki, NDP, NETFLOW, NTP, OSPFv6, PAgP, POP3, PPP, PPPoED, PTP, RADIUS, REP, RIPng, RTP, SCCP, SMTP, SNMP, SSH, STP, SYSLOG, TACACS, TCP, TFTP, Telnet, UDP, USB, VTP
Edit Filters
Show All/None
Event List
Realtime
Simulation
Scenario 0
New
Delete
Fire Last Status Source Destination Type Color Time(sec) Periodic Num Edit
Successful PC0 PC1 ICMP 0.000 N 0 (edit)
Successful PC1 PC0 ICMP 0.004 N 1 (edit)